

Conservation genomics of the endangered Seychelles Magpie-Robin (*Copsychus sechellarum*): a unique insight into the history of a precious endemic bird

EMILY L. CAVILL,^{*1,2}  SHYAM GOPALAKRISHNAN,¹  LARA C. PUETZ,¹  ÂNGELA M. RIBEIRO,³ 
 SARAH S. T. MAK,¹  RUTE R. DA FONSECA,⁴  GEORGE PACHECO,¹  BRONWYN DUNLOP,⁵
 WILNA ACCOUCHE,⁶ NIRMAL SHAH,⁷  ANNA ZORA,⁸ LÍCIA CALABRESE,^{9,10}  MARTIN GENNER,² 
 GARETH JONES,²  CHUNXUE GUO,¹¹ GUOJIE ZHANG^{11,12,13,14} & M. THOMAS P. GILBERT^{1,15} 

¹Centre for Evolutionary Hologenomics, The GLOBE Institute, University of Copenhagen, Copenhagen, Denmark

²School of Biological Sciences, University of Bristol, Bristol, UK

³Natural History Museum of Denmark, University of Copenhagen, Copenhagen, Denmark

⁴Centre for Macroecology, Evolution and Climate, The GLOBE Institute, University of Copenhagen, Copenhagen, Denmark

⁵Cousine Island Company, Victoria, Seychelles

⁶Green Islands Foundation, Victoria, Seychelles

⁷Nature Seychelles, Roche Caiman, Seychelles

⁸Fregate Island Sanctuary Ltd, Victoria, Seychelles

⁹Island Conservation Society, Pointe Larue, Seychelles

¹⁰Island Biodiversity & Conservation Center, University of Seychelles, Anse Royale, Seychelles

¹¹State Key Laboratory of Genetic Resources and Evolution, Kunming Institute of Zoology, Kunming, China

¹²China National Genebank, BGI-Shenzhen, Shenzhen, China

¹³Centre for Excellence in Animal Evolution and Genetics, Chinese Academy of Sciences, Kunming, China

¹⁴Villum Centre for Biodiversity Genomics, University of Copenhagen, Copenhagen, Denmark

¹⁵University Museum, Norwegian University of Science and Technology, Trondheim, Norway

The Seychelles Magpie-Robin *Copsychus sechellarum* is an IUCN Red-List Endangered species endemic to the Seychelles, whose population was reduced to eight individuals on a single island in the 1960s. Translocations from the remaining population to four additional islands have been an integral factor in their recovery, but the potential genetic consequences of their translocation history have not previously been explored. We resequenced the genomes of 141 individuals sampled across the five current island populations and analysed the data to characterize their population structure, as well as to explore suspected inbreeding. Overall, very low levels of heterozygosity were observed, all coupled with long homozygous segments that suggest recent inbreeding, probably the consequence of a population bottleneck in the 1960s. Three of the four translocated populations displayed less genetic diversity than the founder population from which they were taken, a familiar pattern observed as a result of the evolutionary force of genetic drift following founder events. Furthermore, and perhaps surprising given the recent time since the new populations were established, population structure was observed within these same three populations. New awareness of inbreeding in the Seychelles Magpie-Robin populations, and continued genetic monitoring, will allow for genetically informed management decisions. This is particularly prudent in maximizing the success of the future conservation translocation planned for this species.

*Corresponding author.

Email: emily.cavill@live.co.uk

Twitter: @birdiestofnerds